Quantification: 877_wt_z073c1-2
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	14
interior lacuna count	12
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2834.45	2671	1845.67	px^2
major_axis_length_px	lacuna shape	93.8249	101.574	35.0777	px
minor_axis_length_px	lacuna shape	39.1988	36.8708	15.391	px
aspect_ratio	lacuna shape	2.5302	2.1489	1.062	unitless
eccentricity	lacuna shape	0.8781	0.8851	0.077	unitless
solidity	lacuna shape	0.7896	0.7983	0.0629	unitless
roots_count	headline	9.3333	8	7.9582	count
ring_length_r30_px	headline	362.083	351.5	231.526	px
ring_length_r60_px	reported	864.75	831	456.303	px
owned_length_px	ownership-dependent	1614.69	832.302	1704.53	px
edge_count	ownership-dependent	79.3333	37	89.1753	count
mean_edge_length_px	ownership-dependent	22.5199	21.275	12.2739	px
perimeter_px	normalised	276.988	291.519	124.706	px
ring_area_r30_px2	normalised	9249.67	9621	3338.82	px^2
ring_area_r60_px2	normalised	22212.8	22572	7302.53	px^2
in_frame_fraction_r30	normalised	0.9883	1	0.0406	unitless
in_frame_fraction_r60	normalised	0.9695	1	0.0761	unitless
ring_density_r30	normalised	0.0363892	0.0381433	0.0155354	px^-1
ring_density_r60	normalised	0.0368856	0.0382881	0.0124574	px^-1
roots_per_100px_perimeter	normalised	2.843	2.9092	1.6923	per 100 px
ring_attached_length_r30_px	network v2	314.917	308.5	233.878	px
ring_attached_length_r60_px	network v2	644.5	511	510.967	px
ring_length_w_r30_px	network v2	414.144	410.419	264.426	px
ring_length_w_r60_px	network v2	993.627	958.513	527.703	px
sholl_crossings_r10	network v2	8.1667	7.5	5.1669	count
sholl_crossings_r20	network v2	10.5833	9.5	6.8551	count
sholl_crossings_r30	network v2	13.4167	14.5	7.1536	count
loop_count	loops, report only	8.0833	3.5	11.3815	count
loop_edge_share	loops, report only	0.2425	0.2586	0.2086	share
ring_width_mean_r30_px	width	6.7191	6.8224	1.0173	px

Per field
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measure	value	units
analysed_area_px2	1.01608e+06	px^2
total_skeleton_length_px	31505	px
canalicular_length_density_per_px	0.0310065	px^-1
junction_count	668	count
junction_density_per_px2	0.00065743	px^-2
lacuna_count	14	count

lacunae_per_px2	1.378e-05	px^-2
skeleton_component_count	366	count
mean_component_length_px	86.0792	px
median_component_length_px	24	px
field_length_density_w_per_px	0.0355598	px^-1
field_density_without_flagged_per_px	0.031259	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	510	px
unattached_component_count	28	count
canalicular_length_density_attached_per_px	0.0310065	px^-1
field_length_density_with_unattached_per_px	0.0315085	px^-1
loop_count	121	count
loop_edge_share	0.2866	share
edge_count	1640	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4605	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	30049	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	954	58.9902	25.4792	2.3152	0.9019	0.6974	1.3025	49.6237	860.655	False
2	6236	133.392	66.6525	2.0013	0.8662	0.7565	0.7065	193.518	518.807	False
3	1740	108.483	22.6499	4.7895	0.978	0.7716	1.1221	244.43	785.516	False
4	4847	142.75	45.7551	3.1199	0.9472	0.8253	0.4966	381.669	319.923	False
5	2583	119.873	29.7486	4.0295	0.9687	0.7747	0.5429	394.401	41.6202	False
6	4698	104.479	62.6265	1.6683	0.8004	0.7983	0.887	422.75	620.834	False
7	2761	84.1795	44.2264	1.9034	0.8509	0.8577	1.3929	435.645	810.324	False
8	3441	99.3913	46.2527	2.1489	0.8851	0.8428	1.0959	709.919	822.946	False
9	2671	101.574	36.8708	2.7549	0.9318	0.8211	0.7949	796.647	571.938	False
10	654	43.6248	28.1155	1.5516	0.7646	0.6728	1.4274	853.878	586.749	False
11	594	35.3373	22.8099	1.5492	0.7638	0.8672	0.7727	907.293	697.131	False
12	740	61.1371	17.5677	3.4801	0.9578	0.8675	-1.5683	1016.29	147.323	True
13								128.483	610.823	False
14								1017.52	842.011	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	9	351	769	2021.38	80	25.2673	6.043
2	25	793	1609	5317.75	271	19.6227	7.436
3	7	352	989	1841.76	92	20.0191	7.5283
4	19	518	1283	2957.39	145	20.3958	7.6229
5	5	227	592	528.375	28	18.8705	5.7804

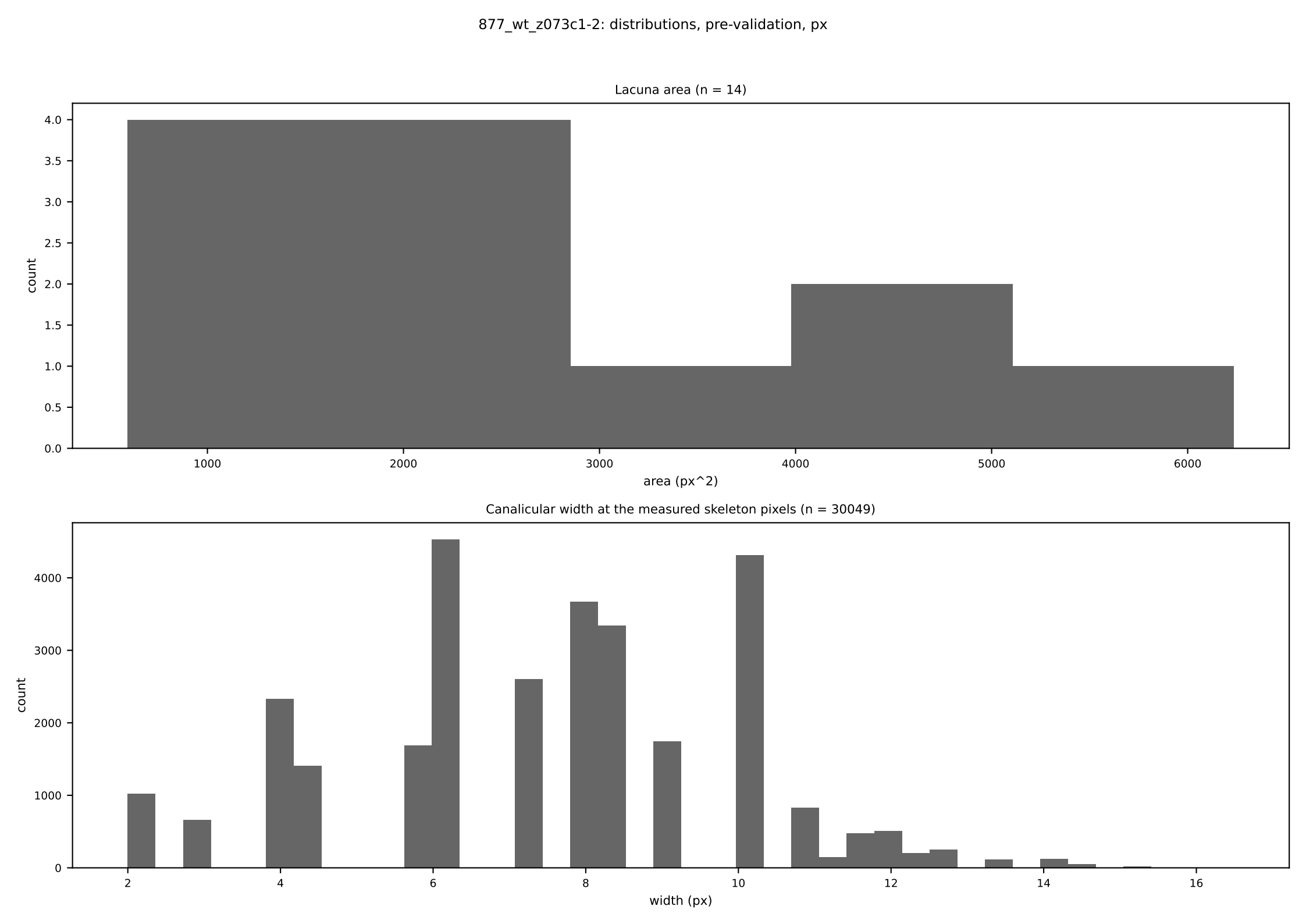
6	19	732	1544	4074.78	222	18.3549	7.324
7	9	361	893	753.24	34	22.1541	7.269
8	11	441	993	601.014	26	23.1159	6.212
9	6	207	689	911.365	40	22.7841	6.3758
10	1	32	134	55.5269	1	55.5269	5.3467
11	0	103	268	0	0	0	5.2483
12	0	0	0	0	0	0	
13	1	228	614	313.664	13	24.128	8.4423
14	0	6	103	0	0	0	3.2168

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	218.244	7625	18201	1	0.8799	0.0460328	0.0422504	4.1238
2	510.192	15054	32669	1	1	0.052677	0.0492516	4.9001
3	294.09	10047	25475	1	1	0.0350353	0.0388224	2.3802
4	387.789	13352	32069	1	1	0.0387957	0.0400075	4.8996
5	312.576	9613	20973	0.8593	0.7542	0.0236139	0.0282268	1.5996
6	397.546	12204	28960	1	1	0.0599803	0.0533149	4.7793
7	261.764	9629	24171	1	1	0.0374909	0.0369451	3.4382
8	305.262	10522	26302	1	1	0.0419122	0.0377538	3.6035
9	288.948	8146	18856	1	1	0.0254112	0.0365401	2.0765
10	164.894	4521	10410	1	1	0.00707808	0.0128722	0.6064
11	124.024	5803	16215	1	1	0.0177494	0.0165279	0
12	159.918	3763	10281	0.5406	0.5269	0	0	0
13	58.5269	4480	12252	1	1	0.0508929	0.0501143	1.7086
14	93.598	3014	8836	0.5506	0.5329	0.00199071	0.0116569	0

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	310	538	406.09	897.477	8	8	16
2	768	1558	893.754	1846.92	15	23	19
3	324	895	415.889	1148.12	8	11	17
4	430	1163	594.571	1493.73	12	15	21
5	157	324	257.108	676.754	7	5	11
6	717	1377	841.311	1772.95	19	22	26
7	346	776	414.747	1019.55	10	11	13
8	307	484	503.274	1134.4	7	13	16
9	147	220	238.35	785.139	4	7	8
10	27	27	32.1421	147.882	3	0	1
11	50	63	113.083	302.664	2	5	5
12	0	0	0	0	0	0	0
13	196	309	259.409	697.943	3	7	8
14	0	0	5.8284	118.054	0	0	1


877_wt_z073c1-2: distributions, pre-validation, px
Lacuna area (n = 14)
count
area (px^2)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
1000
2000
3000
4000
5000
6000
Canalicular width at the measured skeleton pixels (n = 30049)
count
width (px)
0
1000
2000
3000
4000
2
4
6
8
10
12
14
16